Local Projections vs VARs

Topics in Macroeconomics – Week 1

Lukas Boehnert

PhD elective, 2026/27

Bank of Portugal & Católica Lisbon

The Course

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- **Five weeks, three parts:**
  - I) Empirical methods (Weeks 1–2): Local projections vs VARs, shift-share IV
  - II) International macro (Weeks 3–4): Financial crises, open-economy monetary policy
  - III) Macro-development (Week 5): Misallocation, spatial economics

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  - Paper presentation (30%): one paper in the reading group, Weeks 2–4
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$\Rightarrow$ Goal: a project that could become the first draft of a thesis chapter

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Material draws on Mavroeidis (Oxford Macroeconometrics), Plagborg-Møller (2026 LP lectures) and the Montiel Olea et al. (2025) primer.

Why impulse responses?

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- Three questions we ask of the data:
  1. Impulse response: what happens after a policy tightening, month by month?
  2. Variance decomposition: how important are policy shocks for output?
  3. Historical decomposition: which shocks drove the UK recession of the early 1990s?

[Stock & Watson (2018)]

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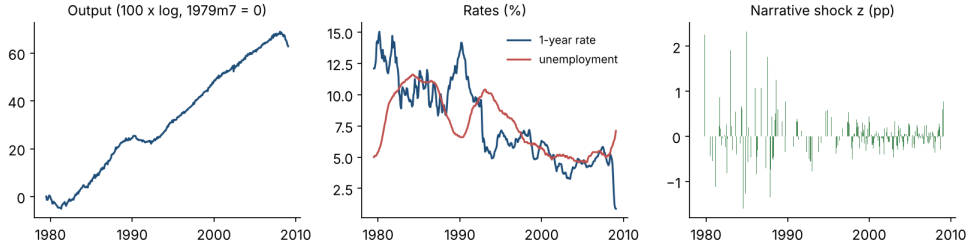
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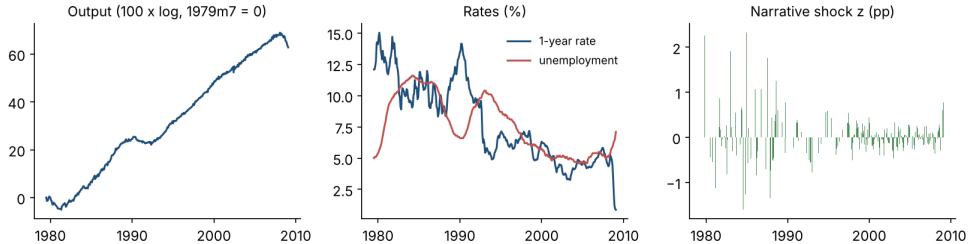
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$\Rightarrow$ Weeks 3–5 lean on these tools too: crises, open-economy policy, regional shocks

Running example: UK monetary policy and output

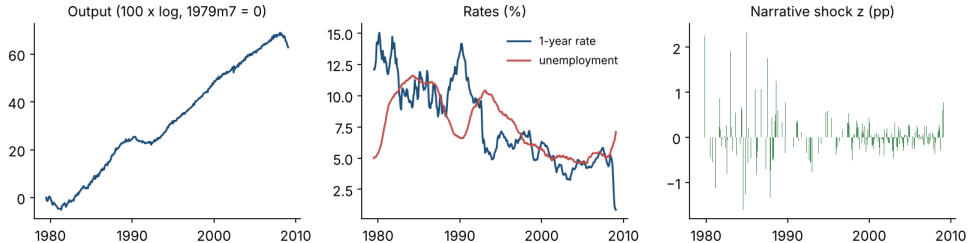


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- Monthly, 1979m7–2009m2, $T = 356$ [data: Cesa-Bianchi, Thwaites & Vicondoa (2020)]
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- **Question:** what does a 1pp higher 1-year rate do to output, month by month?

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- Random rate experiments are infeasible $\Rightarrow$ need either
  - a natural experiment (instrument), or
  - assumptions (timing, signs, ...)

[Mavroeidis, Oxford Macroeconometrics, lecture 4]

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$\Rightarrow$ The LP vs VAR debate is about estimation, not identification [Plagborg-Møller & Wolf (2021)]

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$\Rightarrow$ Every number and figure comes from `week1_simple.py` (one cell per step) and `week1_fast.py`

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- What it is not: a model of the system, or free identification

[Jordà (2005)]

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- y_{t+h}: outcome — UK: output y_{t+h}
- x_t: impulse variable — 1-year rate i_t
- r_t: contemporaneous controls (not moved by the shock within the period) — y_t, u_t
- $w_{t-1}, \ldots, w_{t-p}$: lagged controls; always include lags of y and x — 12 lags of y, u, i
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$\Rightarrow$ y and x come from the question; the controls are the identifying assumption

[Plagborg-Møller (2026), lecture 1]

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- **In practice:** run one regression with all controls (second-step SEs are wrong), but plot $\tilde{x}_t$ and report its s.d.

  Climate example: sd $\approx$ 0.1°C, so "+1°C" is a huge shock [Bilal & Känzig (2025)]

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$\Rightarrow$ Let's start with 1): timing restrictions and adding lags

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Identification by timing restrictions

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$\Rightarrow$ Same logic as a recursive (Cholesky) ordering: we meet it again as the recursive VAR in V1

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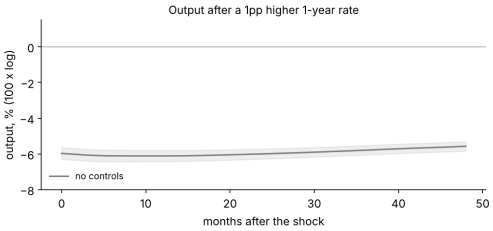
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- **Observed shock?** [e.g. OPEC news surprises, Känzig (2021)]
  - Controls not needed for consistency, but add lags of y, x to shrink SEs
  - If adding lags moves the estimate a lot $\to$ your "shock" is predictable

[Plagborg-Møller (2026), lecture 1]

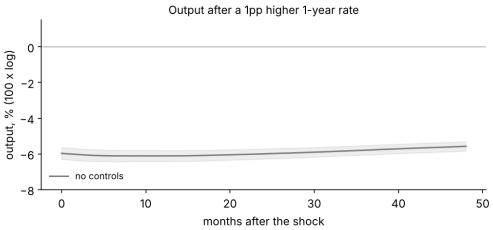
L1: no controls

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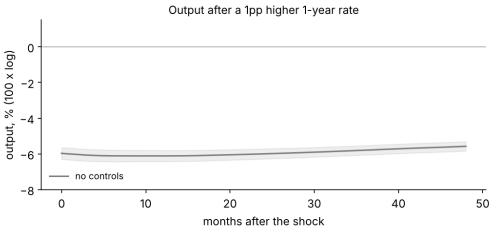
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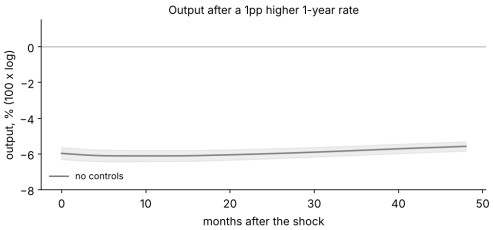
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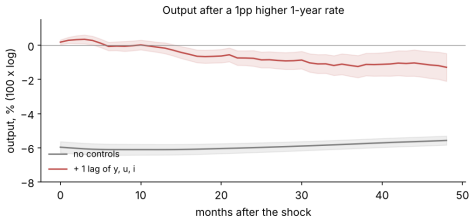


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$\Rightarrow$ $\tilde{x}_t = i_t$: the "shock" is the level of the rate, which nobody believes is random

L2: add lags, then date-t controls

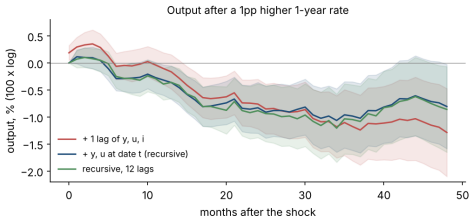
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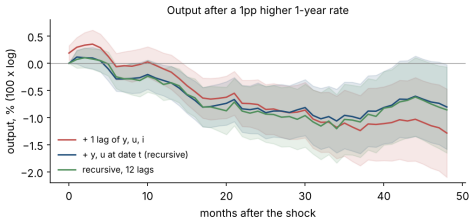
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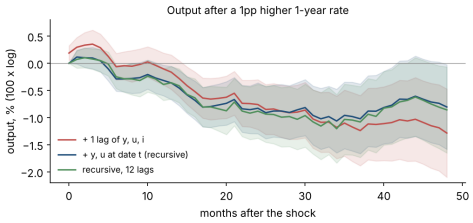
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$\Rightarrow$ Lags remove what is predictable; date-t controls add a timing assumption (V1)

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- Pointwise bands: $\hat{\theta}_h \pm 1 \cdot SE$ (68%), $\pm 1.645 \cdot SE$ (90%); whole-path claims need a sup-t band [Montiel Olea & Plagborg-Møller (2019)]

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- Narrative proxies work too: our z_t reads policy decisions off the Bank's records

[Plagborg-Møller (2026), lecture 1]

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$\Rightarrow$ Pick a normalization variable that moves strongly with z. Proxy only: the FEVD is not identified; using z as if it were the shock understates it

[Plagborg-Møller (2026), lecture 1]

Where instruments come from

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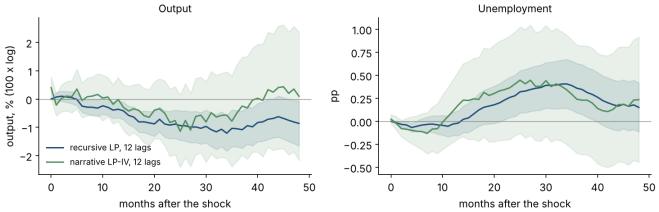
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- **Known threats:**
  - Information effects: surprise reveals central bank's view [Nakamura & Steinsson (2018), Jarociński & Karadi (2020)]
  - Predictability: surprises correlate with past data [Bauer & Swanson (2023), Miranda-Agrippino & Ricco (2021)]

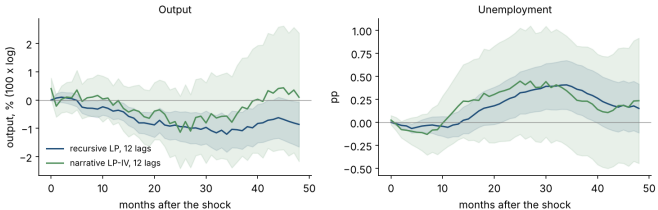
L3: the narrative shock as an instrument

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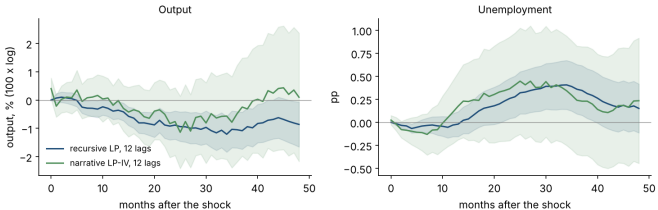
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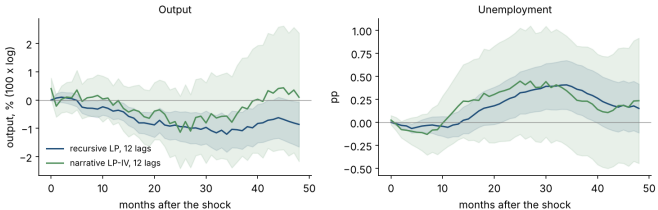
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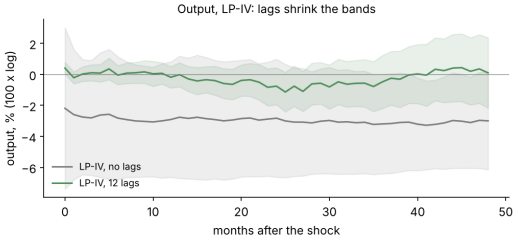


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$\Rightarrow$ A different assumption gives a similar answer at 2 years, with wider, more honest bands

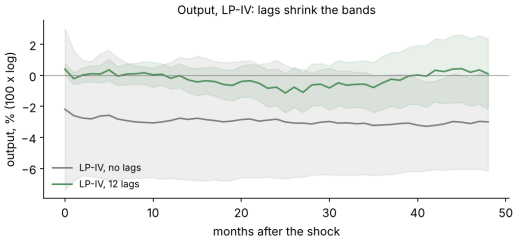
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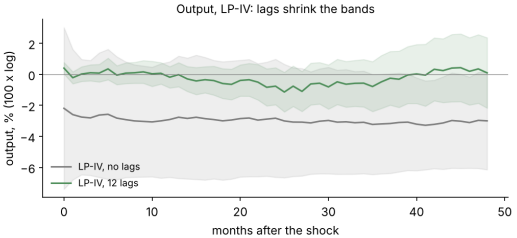
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- No lags: -2.9% at every horizon, se 2.3 $\rightarrow$ the trend again, and 5 times the standard error

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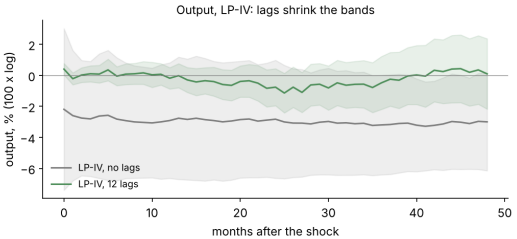
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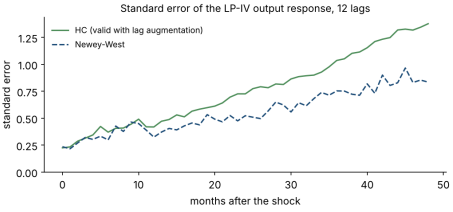


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$\Rightarrow$ Lags soak up residual variance and the predictable part of z_t: always include them

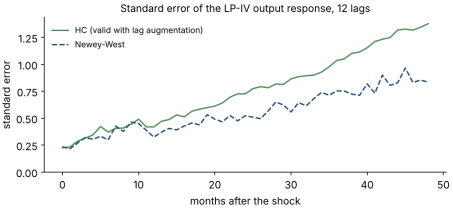
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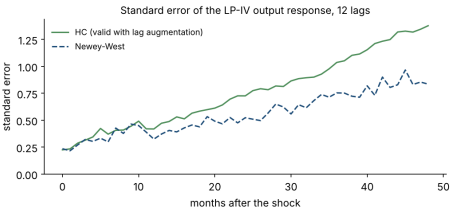
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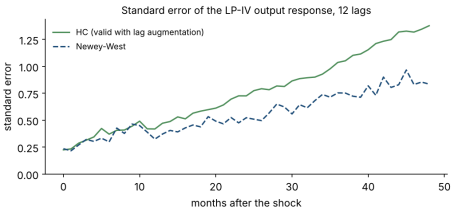
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$\Rightarrow$ With lags as controls, plain HC standard errors are valid: use them

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$\Rightarrow$ A visual aid: always report the band of the original LP, not one around the smoothed line

[Plagborg-Møller (2026), lecture 3]

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- R^2 **method** [Gorodnichenko & Lee (2020)]
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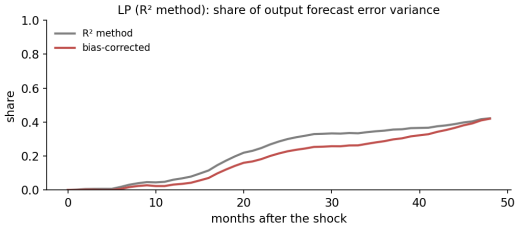
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Running example, $\tilde{x}_t$ from L2, at 12 / 24 / 48 months: raw 5% / 27% / 42%, bias-corrected 2% / 20% / 41%

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$\Rightarrow$ Coming back in Weeks 3–5: crises, open-economy panels, regional LPs

Structural VARs

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$\Rightarrow$ Identification moved from the equations to the shocks – the same frame as our LPs

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$\Rightarrow$ C_h = reduced-form responses to forecast errors: no economic meaning yet

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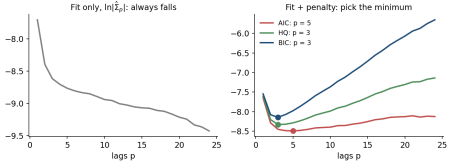
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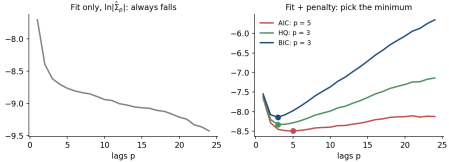
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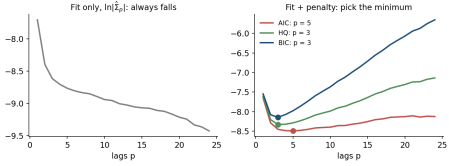


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$\Rightarrow$ ICs are a minimum; with monthly data use at least 12 lags

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$\Rightarrow$ Identification = strip the reaction out of the rate surprise. Simplest way: a regression

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Two variables for now: output and the rate. Regress the rate surprise on the output surprise:

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⇒ Same idea as the LP with date-t controls (L2): the rate, net of this month's output

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$\Rightarrow$ $a = \text{sd}(u_y) = 0.50$, $c = \hat{\gamma} \times a = 0.02$, $d = \text{sd}(e) = 0.36$: exactly the regression

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$\Rightarrow$ UK: the surprises are barely correlated (0.06), so the two nearly agree (V2)

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$\Rightarrow$ **Identification** = supplying the missing restrictions with economics: zeros, an instrument, signs

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- **One restriction closes the gap:**
  - $\beta = 0$: policy affects output only with a lag $\rightarrow$ recursive, rate ordered last
  - Alternatives: γ known, or a sign on β

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$\Rightarrow$ Same assumption as the LP with y_t, u_t as controls (L2): with the same p, both use exactly the same shock [Plagborg-Møller & Wolf (2021)]

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$\Rightarrow$ Bands do not cover bias (too few lags) or a wrong identifying assumption

Three ways to compute VAR bands

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- **Delta method** (formula): unlike in an LP, Θ_h is not a regression coefficient
  - OLS gives SEs for $\hat{A}$ only; translate them to $\Theta_h = g(\hat{A})$ via the slope of g: $\text{Var}(\hat{\Theta}_h) \approx g' \,\text{Var}(\hat{A})\, g'^{\top} \rightarrow$ band $\hat{\Theta}_h \pm 1.645 \cdot SE$
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- **Bayesian** (beliefs): treat A and Σ as random
  - Prior: each series close to a random walk, distant lags close to zero (Minnesota prior)
  - Prior + data $\rightarrow$ posterior; draw (A, Σ), compute responses, take percentiles [Giannone, Lenza & Primiceri (2015)]

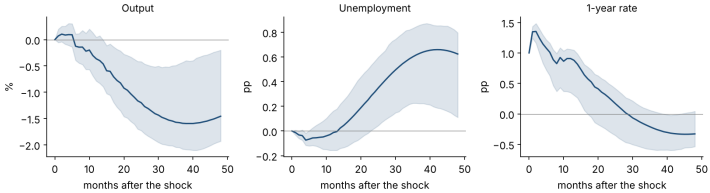
V1: a recursive VAR

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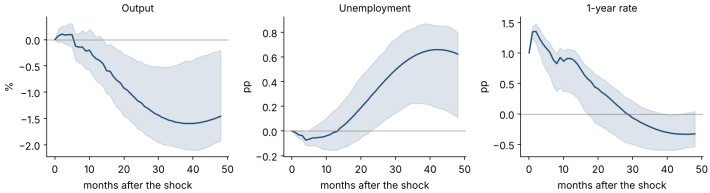
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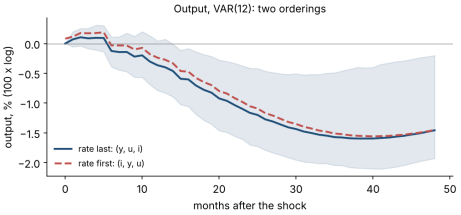


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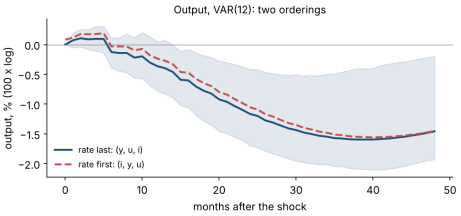
V2: the ordering is the assumption

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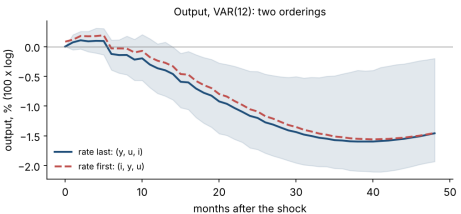
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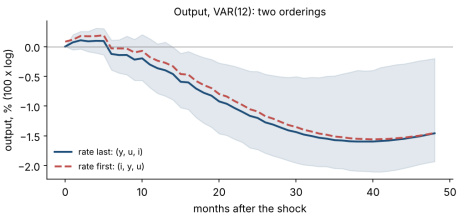
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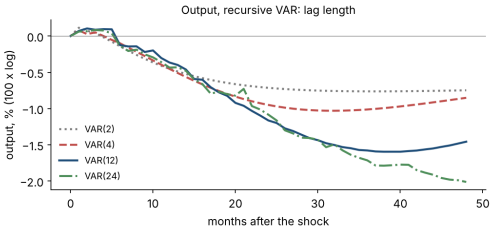


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$\Rightarrow$ With monthly data and few variables, timing assumptions bite less than people fear; with quarterly data, much more

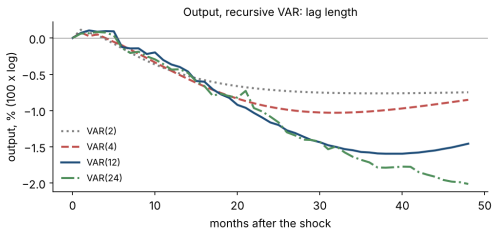
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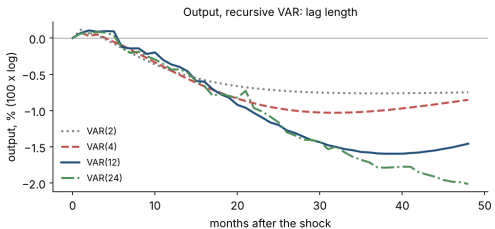
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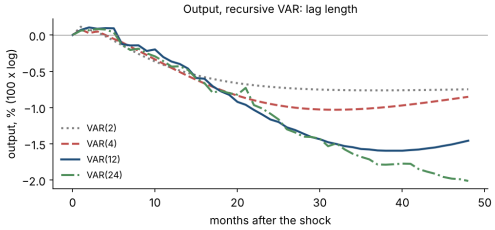
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$\Rightarrow$ Beyond $h = p$ the VAR extrapolates; VAR(24) finds a slow wave (trough at 4 years, back near 0 by 10)

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$\Rightarrow$ UK: output rises on impact ($+0.34$): a warning that z is not entirely clean (V4)

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1) External instrument (SVAR-IV)

the baseline

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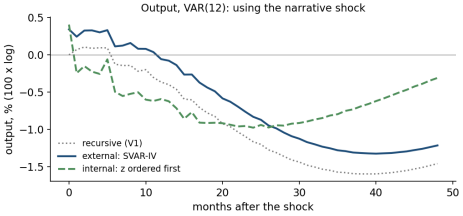
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$\Rightarrow$ If both are valid they agree; if they disagree, trust the internal one (V4)

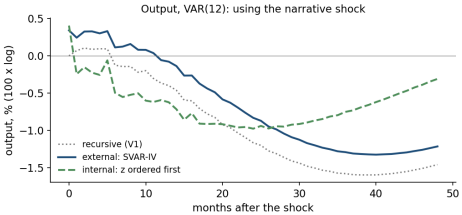
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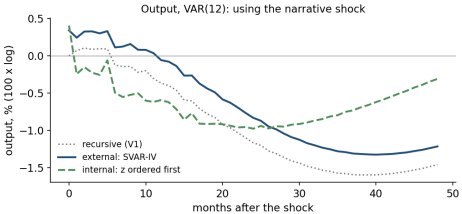
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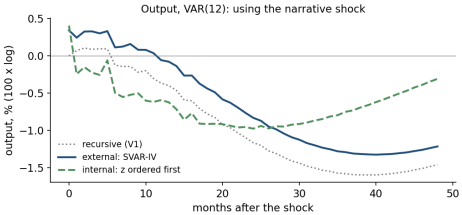
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$\Rightarrow$ All agree on about -1% at 2 years; trust the internal instrument for the shape

Variance and historical decompositions

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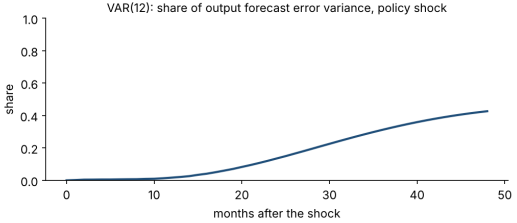
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$\Rightarrow$ Large, significant responses can still be quantitatively minor

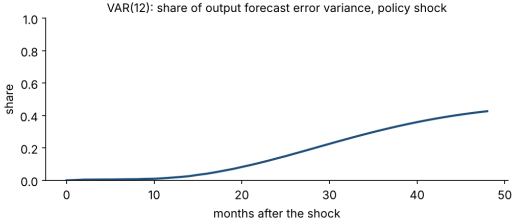
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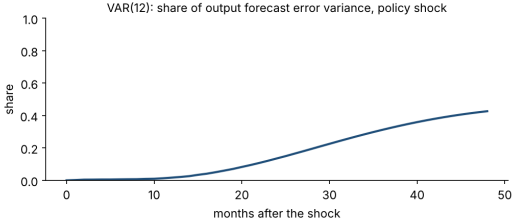
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$\Rightarrow$ Unimportant in the short run, a major driver at business-cycle horizons (if the VAR's extrapolation is right)

A map of identification schemes

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Scheme	Assumption	Reference	Result
Short-run zeros	No response on impact	Christiano et al. (1999)	Point
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- Today in detail: recursive, sign restrictions, external instruments

Other schemes in one slide

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$\Rightarrow$ If you can measure a shock directly (even with noise), do it: more transparent and more robust to nonlinearities [Kolesár & Plagborg-Møller (2025)]

Comparing LPs and VARs

Same estimand

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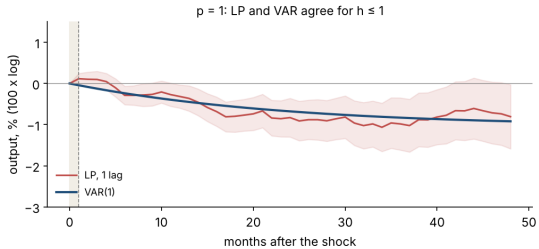
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$\Rightarrow$ Anything you can identify with an SVAR you can identify with an LP, and vice versa

$\Rightarrow$ The choice is about bias and variance in finite samples

C1: same shock, horizon by horizon

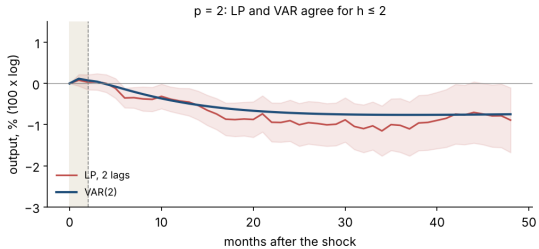
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- Recursive (y, u, i), output response; LP with y_t, u_t and p lags vs VAR(p). Shaded: $h \leq p$
- $p = 1$: the VAR(1) is a smooth curve fitted to one autocovariance

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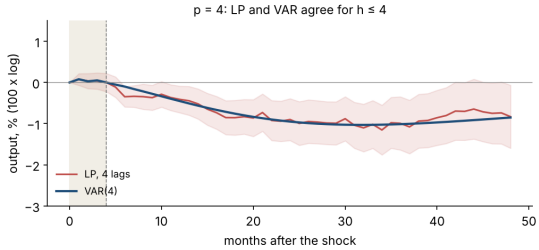
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- $p = 2$: they agree up to $h = 2$; the VAR(2) is too flat afterwards (-0.75 vs -0.89 at 48)

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- $p = 4$: close up to $h = 4$, and here also beyond

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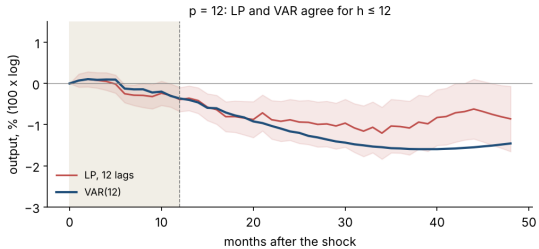
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C1: same shock, horizon by horizon

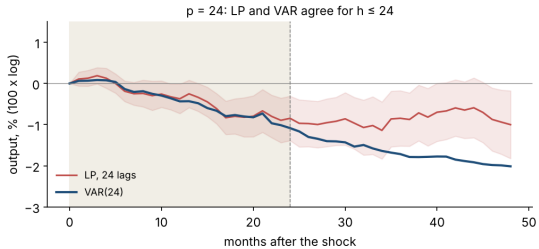
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- $p = 12$: agree up to $h = 12$; at 36 months the VAR(12) says -1.58, the LP -1.05

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- $p = 24$: agree up to $h = 24$; beyond, the VAR(24) keeps falling to -2.0 at 48 months

What the sequence shows

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Lags p	1	2	4	8	12	24
Max $\lvert LP - VAR \rvert$, $h \leq p$	0.16	0.04	0.01	0.06	0.15	0.24
Max $\lvert LP - VAR \rvert$, $h > p$	0.29	0.39	0.27	0.66	0.93	1.32
VAR(p) at $h = 48$	-0.92	-0.75	-0.85	-1.03	-1.46	-2.01
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$\Rightarrow$ Which is closer to the truth is a bias-variance question [Plagborg-Møller (2026), lecture 2]

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Uses z for	Every horizon, via the VAR	Every horizon, directly
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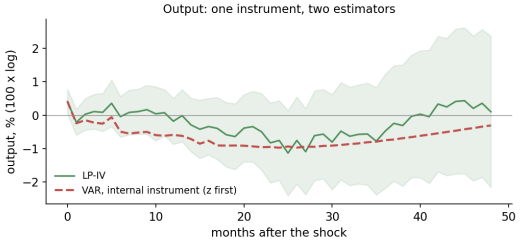
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$\Rightarrow$ Identification is the same; they differ only in estimation: next, bias vs variance

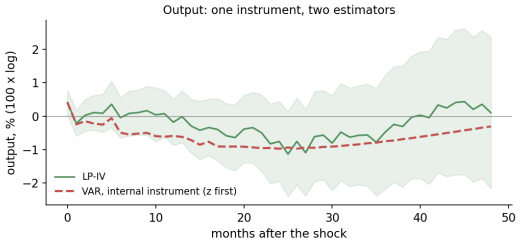
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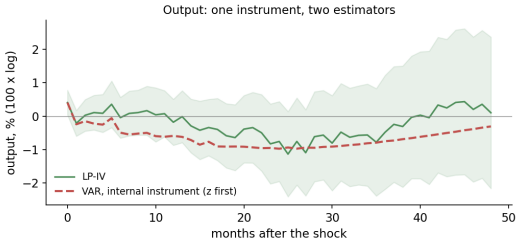
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$\Rightarrow$ Both put output $\sim$0.8–1% lower at 2 years; the VAR is smoother, LP-IV noisier: the trade-off of the next slides

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$\Rightarrow$ Both need the important lags; LP is robust to omitting moderately important ones, the VAR is not

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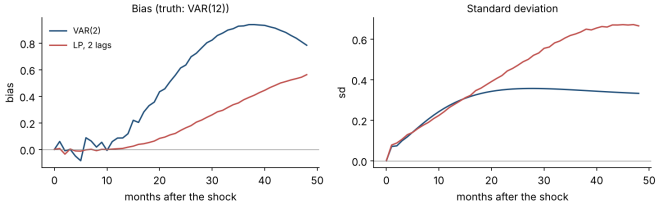
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- **Formally** ("no free lunch"), M = share of MA-residual variance from lagged shocks: [Li, Plagborg-Møller & Wolf (2024)]

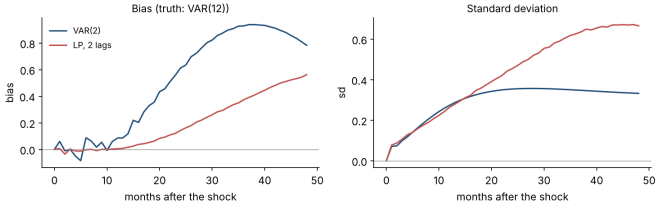
$$|b_h(p)| \leq \sqrt{T \times M \times \left(\tau^2_{h,LP} - \tau^2_{h,VAR}(p)\right)}, \qquad \text{and the bound is attained}$$

$T = 100$, $M = 1\%$, VAR variance half of LP's → worst-case VAR bias = 1.7 standard errors

C3: the trade-off on our data

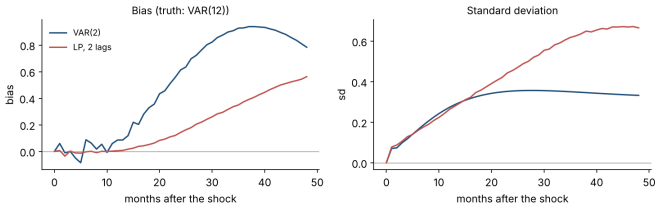


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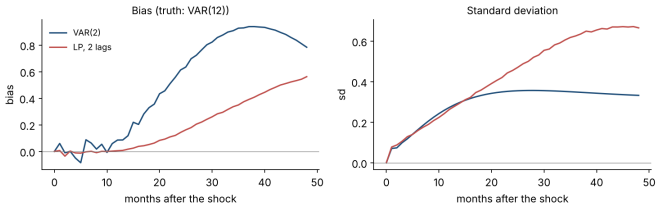
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$\Rightarrow$ VAR bias is twice its sd; LP bias is smaller (not zero: 2 lags do not pin down the true shock) but noisier

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- We want 90% bands that cover for every plausible DGP, not only on average
- A biased VAR band has the right width but is off-centre: with bias $\approx 1.7 \times SE$, a nominal 90% band covers far less
- **Simulation evidence** (200 stationary + 200 non-stationary DGPs, $T = 240$):
  - MSE: (B)VAR with few lags beats LP on average; shrinkage helps both
  - Coverage $\geq$ 80% (target 90%): LP in most DGPs; short VARs fail at medium and long horizons

Uncertainty: bias is costly for confidence bands

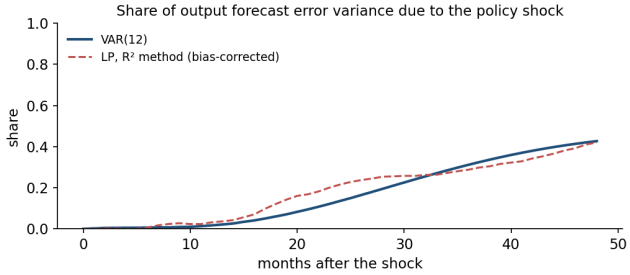
- We want 90% bands that cover for every plausible DGP, not only on average
- A biased VAR band has the right width but is off-centre: with bias $\approx 1.7 \times SE$, a nominal 90% band covers far less
- **Simulation evidence** (200 stationary + 200 non-stationary DGPs, $T = 240$):
  - MSE: (B)VAR with few lags beats LP on average; shrinkage helps both
  - Coverage $\geq$ 80% (target 90%): LP in most DGPs; short VARs fail at medium and long horizons

$\Rightarrow$ Honest bands need an LP, or a VAR with many more lags than usual

[Montiel Olea, Plagborg-Møller, Qian & Wolf (2025)]

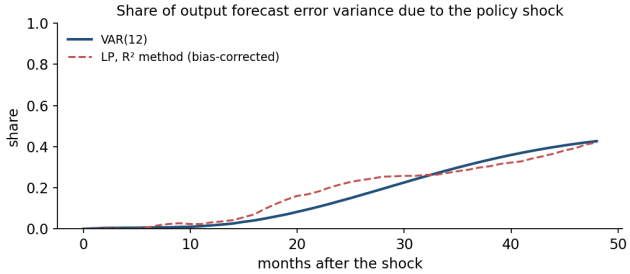
C4: variance decompositions side by side

VAR: $\dfrac{\sum_{\ell<h}[\Theta_\ell]^2_{y,i}}{\sum_{\ell<h}\sum_m[\Theta_\ell]^2_{y,m}}$ (V5) **LP:** R^2 of $\hat{f}_{h,t+h}$ on $\tilde{x}_t, \ldots, \tilde{x}_{t+h}$ (L6)



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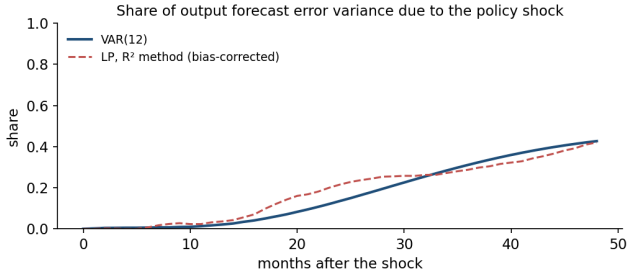
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- 12 / 24 / 48 months: VAR 1% / 12% / 42%; LP (bias-corrected) 2% / 20% / 41%

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$\Rightarrow$ Same shock, same long-run answer; the LP finds policy more important earlier

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- **LP:** linear regression at every h; with lag augmentation the shock is stationary $\to$ inference holds at moderate horizons, even with unit roots
- **But:** LP also has small-sample bias with persistent data ($T \approx 100$): it understates persistence [Herbst & Johannsen (2024)]

Fixes for persistent data

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- **Long differences** with an observed (serially uncorrelated) shock [Piger & Stockwell (2025)]

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⇒ Matters mostly at long horizons with persistent data: run it as a robustness check

[Plagborg-Møller (2026), lecture 3]

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$\Rightarrow$ Depends on your loss function. In a thesis, referees mostly ask whether your bands are honest

[Li, Plagborg-Møller & Wolf (2024), Montiel Olea et al. (2025)]

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7. **Importance:** report a variance decomposition next to the responses

A workflow, start to finish

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Step	What you do	Example
1. Data	Plot everything; levels fine with enough lags [Sims, Stock & Watson (1990)]	L1
2. Lags	AIC on an auxiliary VAR; lean long	L2, L4, V3
3. Shock	Ordering, instrument or signs: state the assumption in one sentence	L3, V1, V2, V4
4. Estimate	SVAR and LP with the same shock and the same p	C1, C2
5. Inference	Bootstrap for the VAR, robust SEs for the lag-augmented LP	L5, V1
6. Importance	Variance decomposition	L6, V5, C4
7. Robustness	More lags, other ordering/instrument, LP vs VAR	V2, V3, C1–C3

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- Next: implementation: the same steps in code

Implementation

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- **Two scripts, same numbers** (`Week1_Lab`):
  - `week1_simple.py`: one cell per step, every LP a regression formula (statsmodels, linearmodels)
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Lab: the running example in code

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  - `week1_simple.py`: one cell per step, every LP a regression formula (statsmodels, linearmodels)
  - `week1_fast.py`: the same cells in fast numpy (all 500 bootstrap samples at once); makes the slide figures
- **In class** ($\sim$25 minutes): run `week1_simple.py` cell by cell: L1–L6, V1–V5, C1–C4
  - Change one thing per step: lags, ordering, instrument, standard errors

LPs in code: a loop over h

```
def ols(Y, X):                                    # drops rows with NaN
    keep = ~np.isnan(np.column_stack([Y, X])).any(axis=1)
    Y, X = Y[keep], X[keep]
    n, k = X.shape
    XXi = np.linalg.inv(X.T @ X)
    b = XXi @ X.T @ Y
    e = Y - X @ b
    V = XXi @ ((X * e[:, None] ** 2).T @ X) @ XXi * n / (n - k)   # HC1
    return b, np.sqrt(np.diag(V)), e, X

# L2(c): output on the rate, controls y_t, u_t and 12 lags of y, u, i
W = np.column_stack([one, y, u, lagged([y, u, i], 12)])
out = np.zeros((H + 1, 2))
for h in range(H + 1):
    b, se, _, _ = ols(lead(y, h), np.column_stack([i, W]))
    out[h] = b[0], se[0]                          # theta_h and its SE
```

$\Rightarrow$ Only the left-hand side shifts across horizons: that is the whole appeal

The VAR in code: estimate once, iterate forward

```
def var_ols(data, p):                          # each equation: OLS on p lags of all
    n = len(data)
    Y = data[p:]
    X = np.column_stack([np.ones(n - p)] + [data[p - l:n - l] for l in range(1, p + 1)])
    B = np.linalg.inv(X.T @ X) @ X.T @ Y
    U = Y - X @ B
    A = [B[1 + (l - 1) * k: 1 + l * k].T for l in range(1, p + 1)]   # A_1..A_p
    return A, U, U.T @ U / (n - p - X.shape[1]), B[0]

def var_irf(A, impact, H):                     # Theta_h = A_1 Theta_{h-1} + ... + A_p Theta_{h-p}
    irf = np.zeros((H + 1, len(impact)))
    irf[0] = impact
    for h in range(1, H + 1):
        for l in range(1, min(h, len(A)) + 1):
            irf[h] += A[l - 1] @ irf[h - l]
    return irf

A, U, Sigma, c = var_ols(np.column_stack([y, u, i]), 12)
P = np.linalg.cholesky(Sigma)                  # recursive: order (y, u, i)
var12 = var_irf(A, P[:, 2] / P[2, 2], H)       # V1: +1pp on the rate
```

Three ways to use the narrative shock

```
# 1. SVAR-IV: VAR residuals on z, then iterate the VAR
zt = z[12:] - z[12:].mean()
slopes = U.T @ zt / (zt @ zt)                    # Cov(u_t, z_t)
svar_iv = var_irf(A, slopes / slopes[2], H)      # +1pp on the rate

# 2. LP-IV (L3): 2SLS of y_{t+h} on i_t, instrument z_t, 12 lags of y, u, i, z
W = np.column_stack([one, lagged([y, u, i, z], 12)])
iv12 = np.array([iv(lead(y, h), i, z, W)[:2] for h in range(H + 1)])

# 3. Internal instrument: recursive VAR in (z, y, u, i), z ordered first
Az, Uz, Sz, _ = var_ols(np.column_stack([z, y, u, i]), 12)
Pz = np.linalg.cholesky(Sz)
svar_int = var_irf(Az, Pz[:, 0] / Pz[3, 0], H)[:, 1:]   # normalise on the rate
```

$\Rightarrow$ 2 and 3 estimate the same object and do not need invertibility; 1 does

Reading group

Gertler & Karadi (2015), AEJ: Macro

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$\Rightarrow$ Template for your presentations: question, contribution, identification, main result, weaknesses, one extension

Reading group and next week

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Rotation

Week	Presents	Discusses
2	[Student A]	[Student B]
3	[Student B]	[Student C]
4	[Student C]	[Student A]
5	Project pitches	All